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Design and Preparation of Low-temperature Curable Multi-permeability Inductor with Low Power Loss for High Frequency Applications

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***Abstract*—This paper proposes a novel low-temperature curable multi-permeability (LTCMP) toroid core fabricated by applying precisely controlled, variable compaction pressure during formation to individual rings requiring a specific permeability. This targeted manufacturing approach enables the permeability of the final core to increase discretely along the radial direction, thereby achieving a significantly more uniform flux density distribution throughout the core structure compared to conventional single-permeability designs. Theoretical electromagnetic calculations and comprehensive finite-element analysis (FEA) simulations were rigorously employed to evaluate and validate the performance characteristics of the multi-permeability cores. Moreover, the inductance behavior and core loss density under specified operating conditions were experimentally characterized and compared for both a representative LTCMP core and a baseline single-permeability core.**

***Index Terms*— Multi-permeability, toroid core, flux density, core loss density, inductance.**

I. Introduction

To enhance the power density and conversion efficiency of a converter, it is necessary to increase the energy density of the magnetic components in the converter [1-6]. Commercial magnetic cores with high permeability are often used to manufacture inductors to obtain high inductance to reduce the size and weight of inductors [7]. However, the magnetic flux of such a toroid core with single-permeability will preferentially saturate on the inner side of the core while the magnetic flux on the outer side remains relatively low, indicating that the flux density distribution within the commercial core is uneven. In fact, the variation of flux density in the toroid core is caused by the variation of the magnetic path. The farther away from the center of the core, the longer the magnetic path, and thus the lower the flux density. For a one-turn commercial toroid core, the flux density at r_x can be expressed as

$$B_x = \mu_0 \mu_r H_{r_x} = \frac{\mu_0 \mu_r I}{l_{r_x}} = \frac{\mu_0 \mu_r I}{2\pi r_x} \quad (1)$$

where μ_0 is the permeability of vacuum, μ_r is the relative permeability of the core, I is the exciting current, l_{r_x} is the length of the magnetic path for any circular cross section of the toroid core, and r_x is the radius of the l_{r_x}.

According to Eq. (1), the flux density is inversely proportional to the r_x of the l_{r_x}. As a result, the low flux density away from the inner side of the core makes part of the magnetic material not fully utilized, resulting in reduced power density and output efficiency of the converters. To make full use of the whole core, it is necessary for the whole core to reach the allowed peak flux density simultaneously to realize uniform flux-density distribution. However, it is difficult to find a way to make the flux density completely uniform in the core. Some compromises [8-16] have been developed to improve the flux density distribution of the core. One method to achieve relatively uniform flux density distribution by changing the external magnetic field of the core from inside to outside. As demonstrated in [8, 10], it consists of three concentric core cells with the same permeability connected by intricate winding. The winding wound in each core cell must be controlled to ensure the external magnetic field increases from the inner side of the core to the outer side. The other design with winding enclosed by the core was subsequently proposed in [9, 11, 12]. It has two halves, and each of which is constructed by inserting a spiral winding into the corresponding slot of the core half. To regulate the magnetic field, the spiral from the inside out has to vary its width and spacing precisely. In conclusion, the intricate winding and geometries of cores undoubtedly increase the complexity of manufacturing.

Compared with the method relying on complicated windings, another method to obtain a uniform flux density distribution by regulating permeability along the radius of the core. Wang *et al*. [13-15] employed low-temperature cofired ceramics (LTCC) technology to fabricate a multi-permeability inductor with ferrite sheets. However, it can be considered as a limited method due to the lack of ferrite sheets with widely distributed permeability as well as the complicated production process, e.g., high-temperature sintering and pressure-assisted lamination

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step [17, 18]. Liu *et al*. [16] used a 3-D printing platform to fabricate multi-permeability cores with different rings. It is more flexible, but a large amount of pastes formulations containing various magnetic components need to be pre-configured to obtain a wide range of permeability to match the definite rings. Besides, the permeability of the printing cores is still in a relatively narrow range on account of the diluted pastes and the press-less processability. The above-mentioned attempt actually divides the core into several regions from the inside out, and each region has a discrete permeability. To address the issue of the discrete distribution of permeability along the radius in the aforementioned multi-permeability cores, we also fabricated the ferrite cores with continuously variable permeability along the radius through gradient sintering [19]. However, the realization of the continuous variable permeability of the core is entirely dependent on the characteristic that the permeability of ferrite materials increases linearly with the rise of sintering temperature. That is to say, this type of core is limited to ferrite materials. In addition, it is very difficult to achieve precise temperature control for different parts of the core during the high-temperature sintering process.

In this paper, we proposed a simple and feasible method to fabricate a low-temperature curable multi-permeability (LTCMP) toroid core using only one magnetic material. The low-temperature curing process is intended to avoid damaging the insulating layer on the surface of the metal particles since the core is an alloy-polymer composite. Usually, this temperature will not exceed 200 °C [20-24]. The complete insulating layer on the surface of the particles can significantly reduce the eddy current losses between the particles, enabling the core to be applied at MHz frequencies. The LTCMP core achieved a gradient distribution of permeability by increasing the compaction pressure from the inner to the outer side of the core. Due to the flexibility of the compaction pressure, the permeability of each ring of the LTCMP core can be adjusted over a wide range without being limited by a single material composition. The preparation of the LTCMP toroid core does not require high-end equipment, such as a 3-D printer or laser cutting machine, which greatly simplifies the manufacturing process of the core. The low-temperature curing process is also conducive to the post-integration of the circuit. In our previous work, the effect of compaction pressure on the permeability of the cores has been studied. The variable permeability obtained under variable compaction pressure can fully meet the desired permeability of the LTCMP core in different regions along the radius.

Section II describes the principle of a LTCMP toroid core. Section III shows the simulation results of LTCMP and single-permeability core, including the flux density distribution, core loss density distribution, and the inductance versus I_{dc}. Section IV describes the procedure of fabricating a LTCMP toroid core and the performance comparison between the two types of cores applied in a DC/DC converter. Section V gives the conclusion.

II. Principle Of LTCMP Toroid Cores

A. Flux Uniformity Factor of the Multi-Permeability Toroid Core

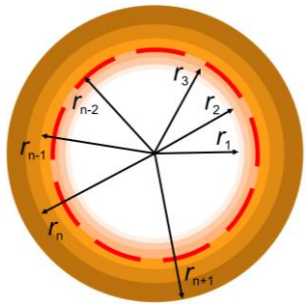


Fig. 1. Structure of a multi-permeability toroid core containing n layers.

Fig. 1 shows the cross-sectional view of the multi-permeability toroid core having the same size as the single-permeability core. It consists of n concentric rings, and the permeability of different rings from the inner side to the outer side of the core increases discretely. The permeability for these rings can be expressed by

$$\mu_{ri} = \begin{cases} \mu_{r1} & r_1 \le r < r_2 \\ \mu_{r2} & r_2 \le r < r_3 \\ \mu_{r3} & r_3 \le r < r_4 \\ \vdots & \\ \mu_{rn} & r_n \le r \le r_{n+1} \end{cases} \tag{2}$$

However, for each individual ring, its flux density is still nonuniform due to the constant permeability. If the maximum and minimum flux density in each ring can be expressed as B_{max} and B_{min}, then the relative permeability of the n rings can be calculated by

$$\mu_{r1} = \frac{2\pi r_1 B_{max}}{\mu_0 I} = \frac{2\pi r_2 B_{min}}{\mu_0 I} \tag{3}$$

$$\mu_{r2} = \frac{2\pi r_2 B_{max}}{\mu_0 I} = \frac{2\pi r_3 B_{min}}{\mu_0 I} \tag{4}$$

$$\mu_{r3} = \frac{2\pi r_3 B_{max}}{\mu_0 I} = \frac{2\pi r_4 B_{min}}{\mu_0 I} \tag{5}$$

$$\vdots$$

$$\mu_{rn} = \frac{2\pi r_n B_{max}}{\mu_0 I} = \frac{2\pi r_{n+1} B_{min}}{\mu_0 I} \tag{6}$$

Combine Eqs. (3), (4), (5), and (6),

$$\frac{B_{min}}{B_{max}} = \frac{r_1}{r_2} = \frac{r_2}{r_3} = \frac{r_3}{r_4} = \cdots = \frac{r_n}{r_{n+1}} = \alpha \tag{7}$$

$$\frac{B_{min}}{B_{max}} = \frac{\mu_{r1}}{\mu_{r2}} = \frac{\mu_{r2}}{\mu_{r3}} = \frac{\mu_{r3}}{\mu_{r4}} = \cdots = \frac{\mu_{rn-1}}{\mu_{rn}} = \alpha \tag{8}$$

where α represents the flux uniformity factor of the multi-permeability toroid core [9, 16, 25]. Obviously, the value of α is $0<\alpha<1$.

Therefore,

$$r_1 = r_2\alpha = r_3\alpha^2 = r_4\alpha^3 = \cdots = r_{n+1}\alpha^n \tag{9}$$

$$\mu_{r1} = \mu_{r2}\alpha = \mu_{r3}\alpha^2 = \cdots = \mu_{rn}\alpha^{n-1} \tag{10}$$

If the number of rings of the multi-permeability core is determined, the α can be easily obtained according to Eq. (9). Then, the permeability of all the rings can be calculated by Eq. (10) as long as the permeability of any one ring is known.

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B. Comparison of inductance for the single-permeability core, multi-permeability core, and constant-flux core of the same size

For a single-permeability toroid core, the permeability of the core along the radius is identical.

$$\mu_{r1} = \mu_{r2} = \mu_{r3} = \cdots = \mu_{rn} \tag{11}$$

The per unit length inductance of the single-permeability inductor with one turn can be expressed by

$$L_{\text{single}} = \int_{r_1}^{r_{n+1}} \frac{\mu_0 \mu_{r1}}{2\pi r} dr \tag{12}$$

After integrating, Eq. (12) is simplified

$$L_{\text{single}} = \frac{\mu_0 \mu_{r1}}{2\pi} \ln \frac{r_{n+1}}{r_1} \tag{13}$$

Then, the inductance of a multi-permeability core can be expressed by

$$L_{\text{multi}} = \sum_{i=1}^{n} \int_{r_i}^{r_{i+1}} \frac{\mu_0 \mu_{ri}}{2\pi r} dr \tag{14}$$

That is to say

$$L_{\text{multi}} = \frac{\mu_0 \mu_{r1}}{2\pi} \int_{r_1}^{r_2} \frac{1}{r} dr + \frac{\mu_0 \mu_{r2}}{2\pi} \int_{r_2}^{r_3} \frac{1}{r} dr + \frac{\mu_0 \mu_{r3}}{2\pi} \int_{r_3}^{r_4} \frac{1}{r} dr + \cdots + \frac{\mu_0 \mu_{rn}}{2\pi} \int_{r_n}^{r_{n+1}} \frac{1}{r} dr \tag{15}$$

After integrating, Eq. (15) is simplified

$$L_{\text{multi}} = \frac{\mu_0 \mu_{r1}}{2\pi} \ln \frac{r_2}{r_1} + \frac{\mu_0 \mu_{r2}}{2\pi} \ln \frac{r_3}{r_2} + \frac{\mu_0 \mu_{r3}}{2\pi} \ln \frac{r_4}{r_3} + \cdots + \frac{\mu_0 \mu_{rn}}{2\pi} \ln \frac{r_{n+1}}{r_n} \tag{16}$$

According to Eq. (9) and Eq. (10), the Eq. (16) can also be expressed by

$$L_{\text{multi}} = \frac{\mu_0 \mu_{r1}}{2\pi} \left(\ln \frac{1}{\alpha} + \frac{1}{\alpha} \ln \frac{1}{\alpha} + \frac{1}{\alpha^2} \ln \frac{1}{\alpha} + \cdots + \frac{1}{\alpha^{n-1}} \ln \frac{1}{\alpha} \right) \tag{17}$$

Simplify Eq. (17) to obtain

$$L_{\text{multi}} = \frac{\mu_0 \mu_{r1}}{2\pi} \ln \frac{1}{\alpha} \left(1 + \frac{1}{\alpha} + \frac{1}{\alpha^2} + \cdots + \frac{1}{\alpha^{n-1}} \right) \tag{18}$$

Dividing Eq. (18) by Eq. (13) to compare the L_{single} and L_{multi}

$$\frac{L_{\text{single}}}{L_{\text{multi}}} = \frac{\ln \frac{1}{\alpha^n}}{\ln \frac{1}{\alpha} \left(1 + \frac{1}{\alpha} + \frac{1}{\alpha^2} + \cdots + \frac{1}{\alpha^{n-1}} \right)} \tag{19}$$

For further calculating

$$\frac{L_{\text{single}}}{L_{\text{multi}}} = \frac{n}{1 + \frac{1}{\alpha} + \frac{1}{\alpha^2} + \cdots + \frac{1}{\alpha^{n-1}}} \tag{20}$$

The inner and outer radii of the previously prepared single-permeability core in [26] were 4 mm and 8 mm respectively. For the multi-permeability core of the same size, if it contains 3, 5, or 10 concentric rings, that is, when n=3, 5, and 10, the corresponding α can be calculated by Eq. (9)

$$\alpha_{3\mu} = \sqrt[3]{\frac{r_1}{r_4}} = \sqrt[3]{\frac{4}{8}} = 0.794 \tag{21}$$

$$\alpha_{5\mu} = \sqrt[5]{\frac{r_1}{r_6}} = \sqrt[5]{\frac{4}{8}} = 0.871 \tag{22}$$

$$\alpha_{10\mu} = \sqrt[10]{\frac{r_1}{r_{11}}} = \sqrt[10]{\frac{4}{8}} = 0.933 \tag{23}$$

Then, the $L_{\text{single}}/L_{\text{multi}}$ can be obtained by substituting α into Eq. (20).

$$\frac{L_{\text{single}}}{L_{3\mu}} = 0.780 \tag{24}$$

$$\frac{L_{\text{single}}}{L_{5\mu}} = 0.744 \tag{25}$$

$$\frac{L_{\text{single}}}{L_{10\mu}} = 0.718 \tag{26}$$

Based on the results of Eqs. (21)-(26), we can infer that the larger the value of n is, the more uniform the flux density distribution of the core will be. Eventually, the ratio of $L_{\text{single}}/L_{\text{multi}}$ will be further reduced. However, the Eq. (18) is applicable for calculating the inductance of a multi-permeability inductor only within the range of $0 < \alpha < 1$. When α=1, the flux density of the core at r_x is constant. For such a constant-flux toroid core, the permeability of the core varies continuously along the radius. Suppose the permeability of the core at $r = r_{n+1}$ is μ_r', then

$$\frac{\mu_{r1}}{r_1} = \frac{\mu_r'}{r_{n+1}} \tag{27}$$

The inductance of the constant-flux inductor can be expressed by

$$L_{\text{multi}*} = \frac{\mu_0 \mu_{r1}}{2\pi r_1} \int_{r_1}^{r_{n+1}} dr \tag{28}$$

After integrating

$$L_{\text{multi}*} = \frac{\mu_0 \mu_{r1}}{2\pi r_1} (r_{n+1} - r_1) \tag{29}$$

Then, the $L_{\text{single}}/L_{\text{multi}*}$ can be expressed by

$$\frac{L_{\text{single}}}{L_{\text{multi}*}} = \frac{\frac{\mu_0 \mu_{r1}}{2\pi} \ln \frac{r_{n+1}}{r_1}}{\frac{\mu_0 \mu_{r1}}{2\pi} \left(\frac{r_{n+1} - r_1}{r_1} \right)} \tag{30}$$

Assuming $r_{n+1}/r_1 = x$, Eq. (30) can be simplified as

$$\frac{L_{\text{single}}}{L_{\text{multi}*}} = \frac{\ln x}{x - 1} \tag{31}$$

When r_1=4 mm, r_{n+1}=8 mm, that is, x =2,

$$\frac{L_{\text{single}}}{L_{\text{multi}*}} = \ln 2 = 0.693 \tag{32}$$

It can be seen that the inductance of a single-permeability core is even less than 70% of that of a constant-flux core of the same size. We are unable to fabricate a constant-flux core to maximize the inductance of the core, but a multi-permeability core of the same size with appropriate permeability layers can still significantly increase the inductance of the core. Dividing Eq. (24) by Eq. (32),

$$\frac{L_{3\mu}}{L_{\text{multi}*}} = 0.888 \tag{33}$$

According to Eq. (33), the inductance of a three-permeability core can account for a considerable proportion of the inductance of the constant-flux core. Considering the complex manufacturing process of multi-permeability core with more permeability layers, the three-permeability core is sufficient.

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III. Flux Density Distribution and Inductance Simulation

In this section, 2D models were established to simulate the distribution of flux density and core loss density of single-permeability core and multi-permeability core, as well as the variation of inductance versus I_{dc}. All the models in the simulation are of the same size, that is, r_{in}=4 mm, r_{out}=8 mm, h=1.8 mm. The relative permeability of the single-permeability core (1-μ) was set to 21.1, which was consistent with that of the core prepared under a pressure of 200 MPa. The radius and relative permeability of each ring from the inside out of the designed multi-permeability core (3-μ, 5-μ, and 10-μ) follow Eq. (8) and Eq. (9).

A. Flux Density Distribution

The 1-μ core, 3-μ core, 5-μ core, and 10-μ core in Figs. 2(a), (b), (c), and (d) show the flux density distribution of single-permeability core and multi-permeability core under 1 A excitation. For the 1-μ core, the flux density on the inner side of the core reached 0.0105 T while only 0.0049 T on the outer side. This result indicated the non-uniformity of the flux density distribution from the inside to the outside of a single-permeability core, resulting in the waste of some core materials. For the 3-μ core, the difference in flux density between the inner (0.0105 T) and outer (0.0078 T) sides of the core was significantly improved due to the discretely variable permeability along the radius. As the core was divided into more regions, i.e., 5-μ core and 10-μ core, the difference had further reduced. More permeability layers mean a greater α, the flux density distribution in each region will of course be more uniform. When the permeability of the core varies continuously along the radius, the flux density in the core can theoretically reach complete uniformity. Obviously, multi-permeability cores enable the whole cores to be utilized more fully.

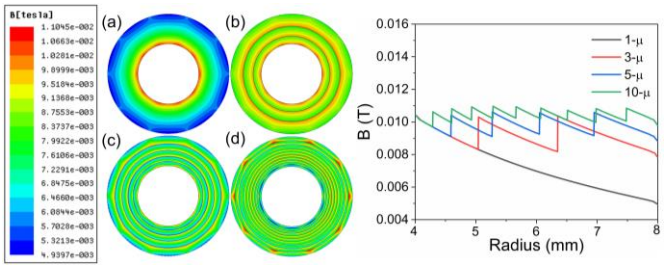


Fig. 2. Flux density distribution of (a) 1-μ core, (b) 3-μ core, (c) 5-μ core, and (d) 10-μ core.

B. Core Loss Density Distributions

The simulated flux density distribution of the cores containing different numbers of permeability layers mentioned earlier has proved the advantages of multi-permeability cores in the utilization of core materials. Since this paper aims to study the preparation and characterization of three-permeability cores, the subsequent simulations mainly focus on comparing single-permeability core with three-permeability core. To simulate the actual working conditions of the cores, a 1 MHz sinusoidal wave with an amplitude of 1 A is used as the current excitation. At this frequency, it is calculated that the skin depth of both single-permeability and three-permeability cores is much greater than the core size. Therefore, the skin effect of the cores can be ignored. The Steinmetz parameters of the single-permeability core and the ring cells of the three-permeability core were all obtained from the software through least squares fitting after importing the B-P curve of the core. The core loss density distribution of the single-permeability and the three-permeability cores is shown in Fig. 3. For the 1-μ core, the core loss density on the inner side of the core is nearly five times that on the outer side. For the 3-μ core, this difference is less than twice in each permeability layer. Obviously, the 3-μ core has a more uniform core loss density distribution than that of a single-permeability core.

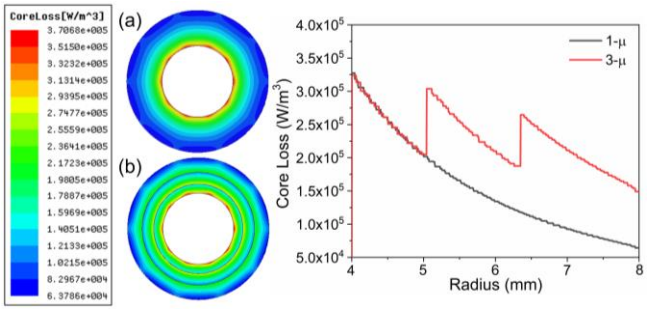


Fig. 3. Core loss density distribution of (a) 1-μ core and (b) 3-μ core.

C. Inductance Versus DC

Fig. 4 shows the simulated L-I_{dc} plots of the single-permeability and three-permeability inductors. Throughout the entire range of I_{dc} bias, the inductance of the three-permeability inductor is greater than that of the single-permeability inductor.

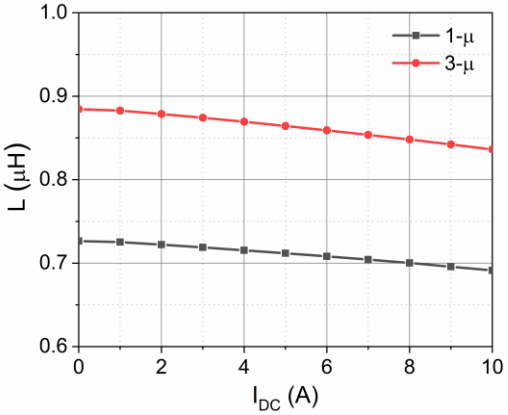


Fig. 4. Simulated inductance of a single-permeability and a three-permeability cores versus I_{dc}.

IV. Prototype Fabrication and Experiment

A. Fabricate a LTCMP Toroid Core

LTCMP toroid cores were fabricated in this work. The core is composed of permalloy powder and TMPTA, and all these raw materials were bought from commercial vendors. The powder and the TMPTA were supplied by Advanced Technology & Materials Co., Ltd and Changxing Chemical Industry Co. Ltd, respectively. The process of mixing the raw

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materials refers to [26]. In our previous work [26], we had obtained the permeability of the cores under different pressures. The average permeability of the samples at 200 MPa, 300 MPa, and 400 MPa was 21.1, 26.2, and 32.9, respectively. When the permeability at 200 MPa is taken as the permeability of the inner ring of the designed LTCMP core, the theoretical permeability of the middle ring and the outer ring are 26.6 and 33.5 according to Eq. (10). It turns out that applying compression pressures of 200 MPa, 300 MPa, and 400 MPa respectively to the inner ring, middle ring, and outer ring of the three-permeability core can roughly obtain the desired permeability of the three rings. The inner and outer radii of each ring calculated according to Eq. (9) are shown in Fig. 5.

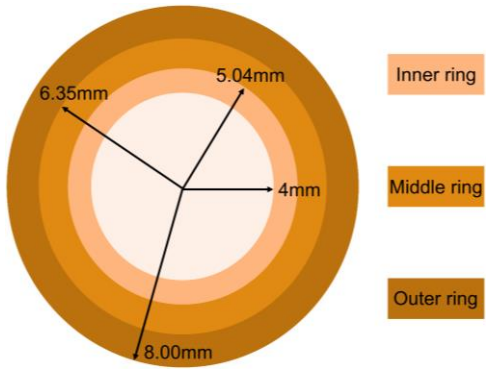


Fig. 5. A cross-sectional view of an LTCMP toroid core.

Fig. 6 is the schematic diagram showing how to fabricate a LTCMP core. The LTCMP core was pressed using a self-designed mold. All the upper punches, under punches, and mandrels of this mold are combinable components to prepare different rings of the LTCMP core. To more clearly demonstrate the pressing process of the core, the mandrels of each ring are not shown in Fig. 7. The LTCMP core is pressed from the outside to the inside. The outer ring is pressed first. After the outer ring is pressed, the upper punch of the outer ring does not need to be removed. Then push out the mandrel of the outer ring, insert the corresponding under punch and mandrel of the middle ring into the under punch of the outer ring. Fill the coated powder into the upper punch of the outer ring, and press it with the upper punch of the middle ring. The final step is to press the inner ring of the core, a process similar to pressing the middle ring. That is, push out the mandrel in the middle ring and then insert the corresponding under punch and mandrel of the inner ring into the under punch of the middle ring, continue to fill in the powder into the upper punch of the middle ring, and press it with the upper punch corresponding to the inner ring to obtain the inner ring. The holding time for each ring is 3 minutes, which is the same as that for pressing single-permeability cores. After the pressing of the three rings is completed, the core is demolded as a whole. Fig. 7 shows the curing curve as well as the cured single-permeability core and the LTCMP core.

The coated powder required for each ring is converted based on its area. Due to the different compression pressures exerted on each ring, this will cause differences in the final thickness of each ring. In order to ensure that the thickness of each ring is uniform, we make the thickness of the inner ring and the middle ring the same as that of the outer ring. That is, when pressing the middle ring and the inner ring, we appropriately reduce the amount of the pre-calculated powder to make its thickness as consistent as possible with that of the outer ring. It should be worth noting that there are gaps left between each ring. These gaps are the mating clearances between the upper punch and the upper punch, as well as between the lower punch and the lower punch. Therefore, this will inevitably cause low-density zones between rings during the pressing process, thereby reducing the inductance of the LTCMP core to a certain extent.

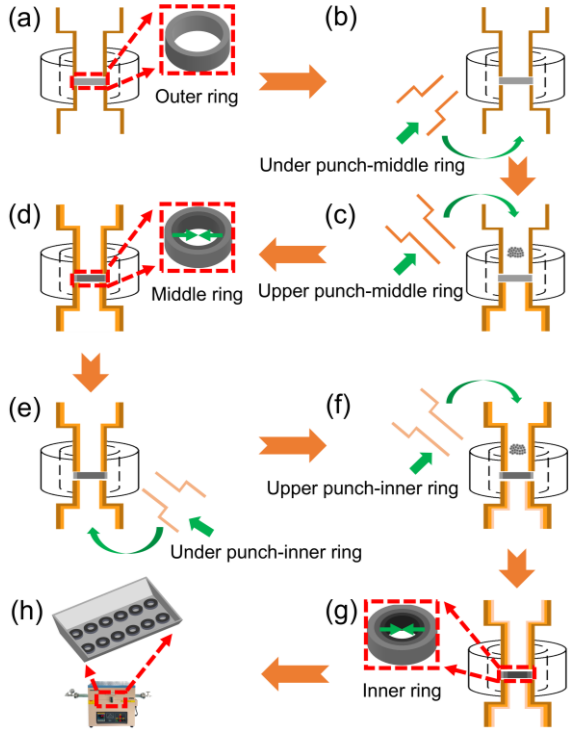


Fig. 6. Process of fabricating a LTCMP toroid core.

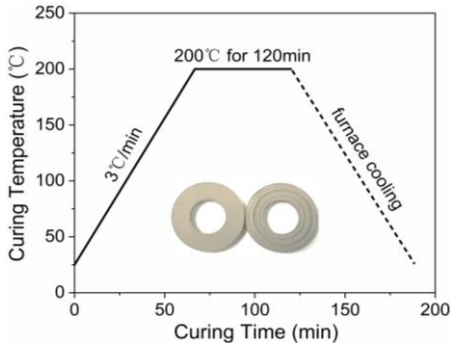


Fig. 7. Cured curve and prototype of a single-permeability and LTCMP cores.

B. Experimental Results

1) Permeability and Inductance Versus DC Current

The relative permeability of the cores was measured by an Impedance Analyzer (E4990A, Agilent) in the frequency range from 100 KHz to 100 MHz. The inductance versus I_{dc} bias at 1 MHz was also characterized using the same impedance analyzer. The cores were wire-wound with ten turns of copper

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winding, and the excitation current varied from 0 to 20 A.

Figs. 8(a) and 8(b) show that the relative permeability of single-permeability and the LTCMP cores has excellent reproducibility. To ensure the accuracy of the data as much as possible, we use samples with relative permeability at an intermediate value as the test objects, although the difference between them can almost be ignored. The plot of the relative permeability of single-permeability and the LTCMP cores is shown in Fig. 8(c). The LTCMP core has a higher relative permeability due to the gradient distribution of the compression pressure, while its frequency stability is comparable to that of a single-permeability core. Fig. 8(d) shows the L-I_{dc} plot of single-permeability and the LTCMP core. The inductance of both gradually decreases as the load increases. Over the entire load range, the inductance of the LTCMP core is more than 20% higher than that of a single-permeability core. The LTCMP core enables the magnetic material far from the inner side of the core to be fully utilized, thus resulting in a higher inductance. Compared with the simulation results in Fig. 5, the measured inductance of the LTCMP core is slightly smaller. It was likely that the dimensional deviation and the inherent defect of the LTCMP core itself during the prototype fabrication, such as micro-cracks or the air gaps between the adjacent rings, which could reduce the overall permeability of the core as well as the inductance of the LTCMP core.

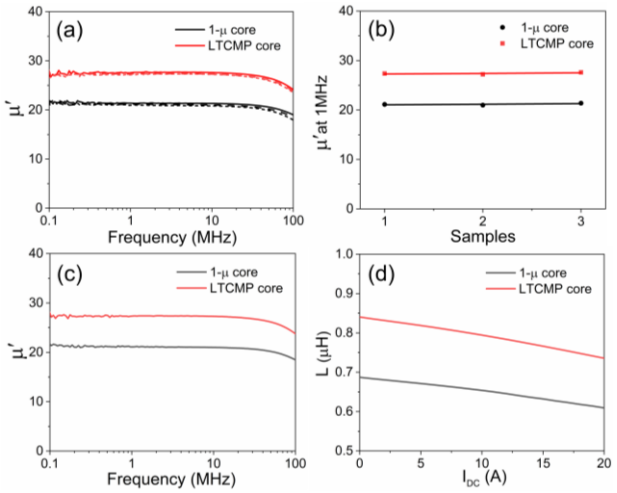


Fig. 8. (a) Reproducibility of relative permeability of single-permeability and the designed LTCMP cores versus frequency. (b) Error analysis in statistics. (c) Relative permeability of single-permeability and LTCMP cores at the intermediate value and (d) the corresponding inductance versus I_{dc} at 1 MHz.

2) Core-Loss Density

The core-loss densities of single-permeability and the LTCMP cores were measured under different conditions using a B-H analyzer (IWATSU, SY8218). Fig. 9(a) shows the plots of core-loss density of both the single-permeability and LTCMP cores at a frequency of 1 MHz within the range of 1~20 mT. The results show that the LTCMP core has 20%~30% lower core-loss density than that of the single-permeability core. To evaluate the applicability under practical converter, the extended loss characteristics from 0.1 to 7 MHz under 5 mT were also measured, as shown in Fig. 9(b). The results show that the prepared LTCMP core still possesses lower core loss density over a wide frequency range. Obviously, this is attributed to the fact that the gradient distribution of pressure in the LTCMP core reduces the total loss of the core.

To explore the influence of compression pressure on the loss of cores, we measured the core loss density of the cores pressed under 200-400 MPa and explained the causes by analyzing the fracture surface morphology and physical properties of these cores, as shown in Figs. 10(a) to 10(i). When the core is pressed under 200 MPa, its particle packing is relatively loose and the density is low due to the low pressure. There are numerous air gaps and non-magnetic resins between the particles, which is equivalent to introducing a large number of distributed air gaps into the core. These air gaps will strongly impede the movement of magnetic domain walls, making the magnetization process difficult and uneven. As a result, the hysteresis loss of the core is relatively large, while the eddy current loss of the core is not significant due to the high resistivity caused by low pressure. When the cores are pressed under 300 MPa and 400 MPa, the particles are tightly pressed together due to the higher pressure, resulting in a higher density and a denser internal structure. Although the eddy current loss of the cores will increase, the stronger magnetic coupling between particles makes the magnetization process easier and leads to a rapid decrease in hysteresis loss. The reduced hysteresis loss exceeded the increase in eddy current loss. Thus, the core loss density of the cores pressed under 300 MPa and 400 MPa is lower than that of the core pressed under 200 MPa.

For the LTCMP core, the core loss of the inner ring is the same as that of the corresponding region of the single-permeability core, but the core losses of the middle and outer rings are both lower than those of a single-permeability core in the same region. This is the reason why the loss density of the LTCMP core is lower than that of the single-permeability core. Furthermore, we compared the loss density of the LTCMP core at 1 MHz with that of the multi-permeability core fabricated by 3D printing in [16], as shown in Fig. 11. The result shows that the core loss density of the designed LTCMP core in this paper is lower.

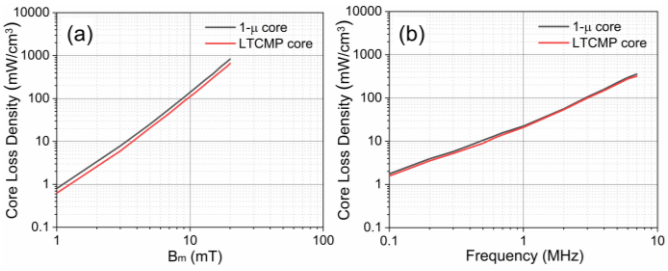


Fig. 9. (a) Core-loss density versus B_m of single-permeability and the LTCMP cores measured at 1 MHz; (b) Core-loss density versus frequency of single-permeability and the LTCMP cores measured under 5 mT.

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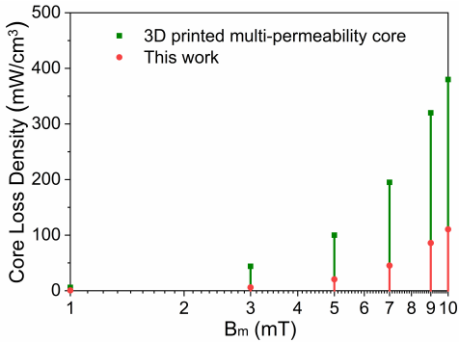


Fig. 11. Comparison of the core-loss density between the LTCMP core and the multi-permeability core in [16].

3) Efficiency in a DC-DC Converter

A 1-MHz 8-V input 1-V output buck converter was constructed to test the two core prototypes. The results were shown in Fig. 12. The efficiency of the converter with the LTCMP core is more than 1% higher than that of the single-permeability core, which means that the prepared LTCMP core can help to improve the output efficiency of the converter.

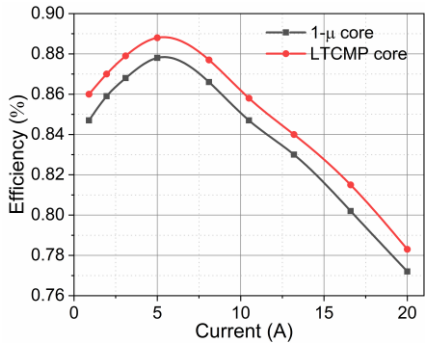


Fig.12. Efficiency of the DC-DC converter with the single-permeability and LTCMP cores.

4) Comparison of Core Loss and Winding Loss

To investigate the loss of inductors, we simulated the core loss and winding loss of both the single-permeability inductor and the designed three-permeability inductor under 1 A, 1 MHz current excitation. The results are shown in Fig. 13. During the steady-state period, for example from 1 μs to 2 μs, the average core losses of the single-permeability inductor and the three-permeability inductor are 21.05 mW and 34.54 mW respectively, while their average winding losses are the same, both being 3.79 mW.

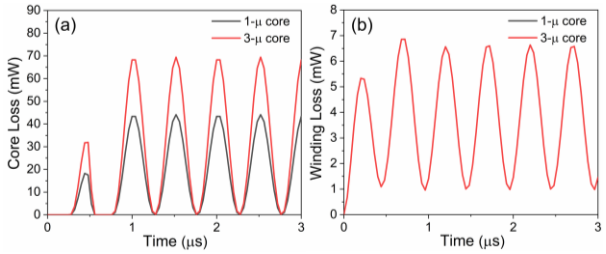


Fig. 13. Simulated core loss and winding loss of the single-permeability and three-permeability inductors: (a) core loss; (b) winding loss.

5) Spatial Progression of Saturation Across Different Permeability Regions

According to (34), the maximum flux densities of the single-permeability cores corresponding to inner ring, middle ring and outer ring of the LTCMP core under a 20 A excitation are calculated as 0.17 T, 0.20 T and 0.22 T, which are far below the saturation flux density of Permalloy. Therefore, we believe that the LTCMP core still operates within the linear range under 0 to 20 A excitation. That is to say, the inner ring, middle ring and outer ring of the LTCMP core will not experience saturation. Based on the above analysis, the distribution of flux density along the radius of the LTCMP core under 0~20 A excitation is shown in Fig. 14.

$$B = \frac{LI}{NA_e} \tag{34}$$

where B is the flux density of the single-permeability core corresponding to each ring of the LTCMP core under different current excitation; I is the excitation current; N is the number of turns in the winding; A_e is the cross-sectional area of the core.

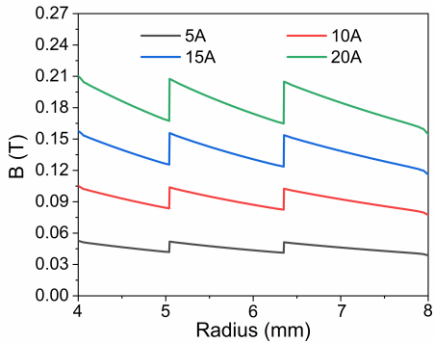


Fig.14. The flux density distribution along the radius of the LTCMP core under 0~20 A excitation.

V. Conclusions

This paper proposed LTCMP toroid cores to improve the flux density distribution in the core by the method of gradient pressurization to achieve full utilization of core materials from the inside out. The results show that the inductance of the LTCMP core under heavy load is at least 20% higher than that of a single-permeability core, while its core loss density is about 20%~30% lower than that of the single-permeability core within the range of 1 to 20 mT. The characterization of the LTCMP core and single-permeability core demonstrate the advantages of multi-permeability core in converter applications. In future, the dimensional deviation and the inherent defect of the LTCMP core itself during the prototype fabrication, such as micro-cracks or the air gaps between the adjacent rings, should be improved to avoid reducing the overall permeability of the core as well as the inductance of the LTCMP core. Furthermore, it'd be better to fabricate the LTCMP cores with more rings with different permeability to further enhance the effective inductance. Of course, the temperature dependence of permeability and loss and the thermo-mechanical reliability of such LTCMP cores should be verified as well.

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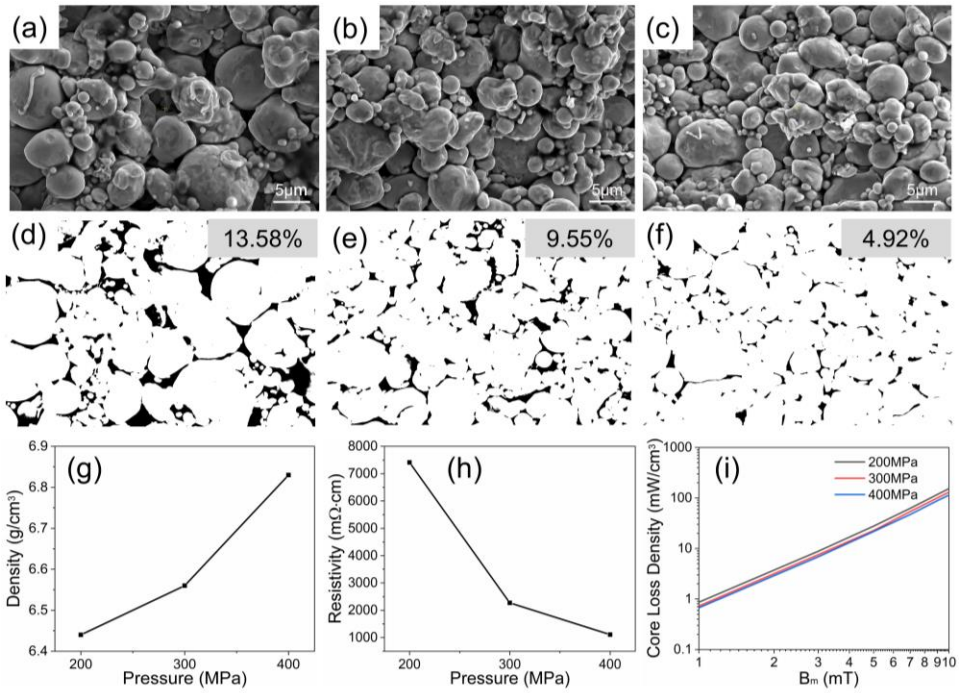


Fig. 10. Fracture surface morphologies of the cores pressed under different pressures: (a) 200 MPa; (b) 300 MPa; (c) 400 MPa; porosity corresponding to the fracture morphologies of each ring: (d) 200 MPa; (e) 300 MPa; (f) 400 MPa; (g) density, (h) resistivity, and (i) core loss density of cores under different pressures.

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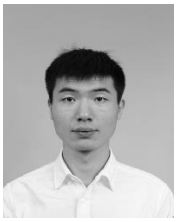
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